

Physiological and biochemical metabolism of germinating broccoli seeds and sprouts.

Changes in physiological and biochemical metabolism as well as glucoraphanin and sulforaphane contents of germinating broccoli seeds and sprouts were investigated in this study. Sprout length, root length, and fresh weight increased with germination time. Dry weight varied from 2.5 to 3.0 mg per sprout. A rapid increase in respiratory rate of sprouts occurred between 24 and 36 h of germination and then stayed at a high level. HPLC analysis found that glucoraphanin content increased at the early stage (0-12 h) of germination, decreased to a low value of 3.02 mg/g at 48 h, and then reached the highest value of 6.30 mg/g at 72 h of germination. Sulforaphane content decreased dramatically during the first day of germination, then increased slowly, and reached a high value of 3.38 mg/g at 48 h before declining again.